Analysis of Higher Order Mode Signals as a Diagnostic for Beam Offset Measurements

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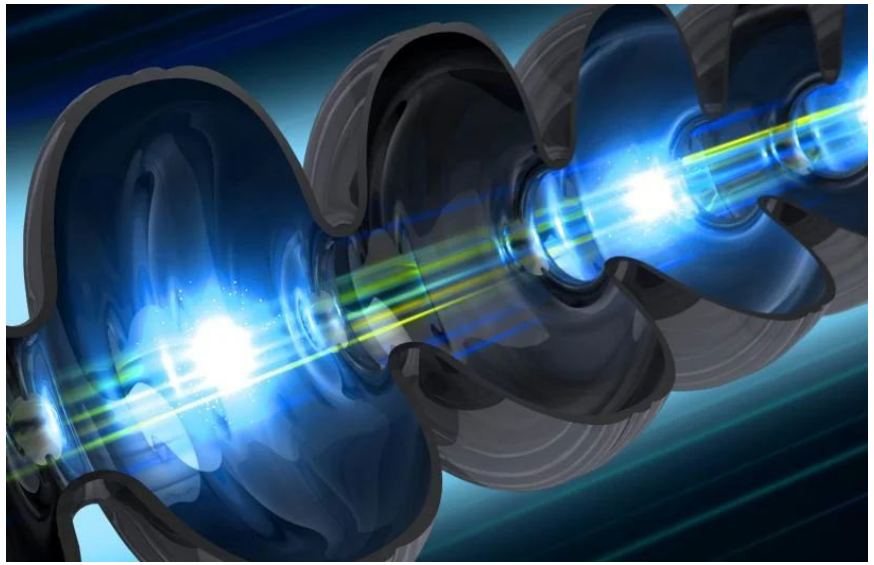
SLAC NATIONAL ACCELERATOR LABORATORY

Stanford University | U.S. DEPARTMENT OF ENERGY

Overview



https://www6.slac.stanford.edu/news/2016-04-04-major-upgrade-will-boost-power-worlds-brightest-x-ray-laser

LCLS-II

What is the LCLS-II and how are experiments carried out?

Higher Order Modes

What are they, and how do they impact the electron beam?

Motivation and Strategy

How can Higher Order Modes be used as a Diagnostic for Beam Offset Measurements?

Software Development

What are the software tools at our disposal to conduct this research?

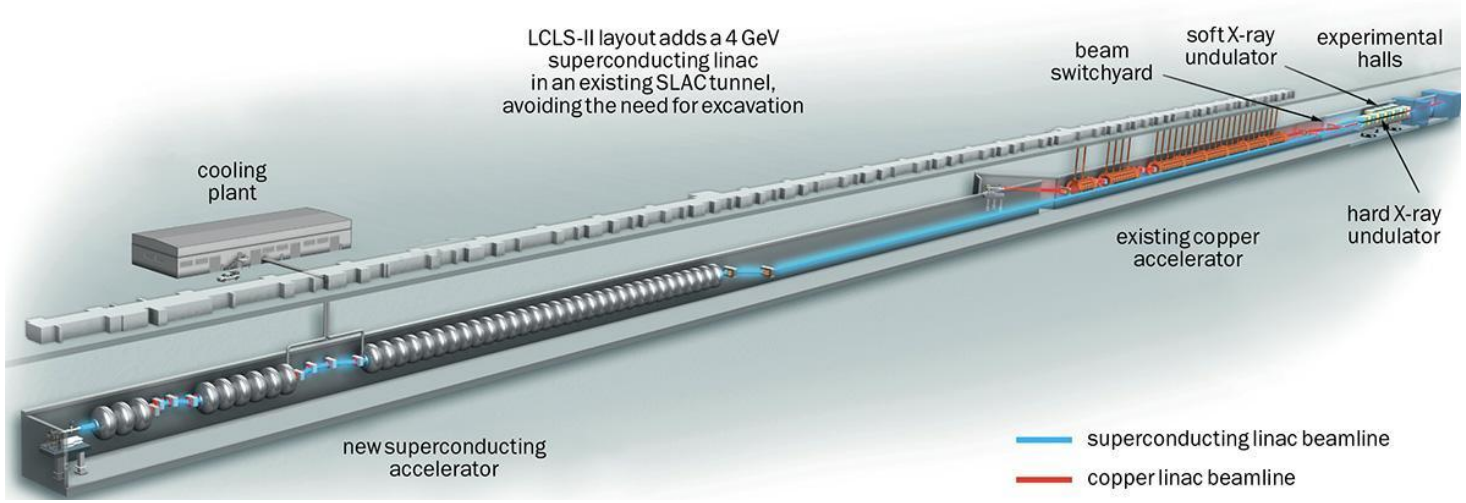
Data Analysis

How do we approach this problem using the data we have? What are new approaches to obtain and process this data?

Conclusions

How can we improve the models we have developed and create an algorithm that gives seamless control over them beam?

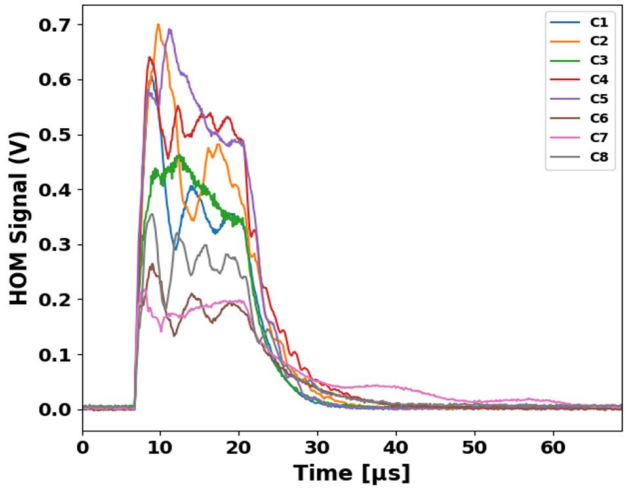
LCLS-II



Overview

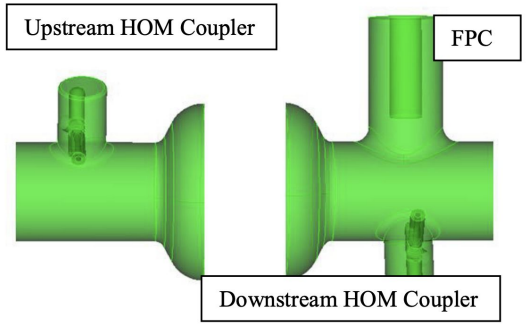
- Electron Beam Accelerator.
- Initial section superconducting, enables high repetition rate of x-ray pulses.
- At frontend, electrons pass through Niobium accelerating cavities cooled by liquid Helium to 2K.
- Radio Frequency (RF) signals are pumped into cavities by Solid State Amplifiers (SSA) that add energy and accelerate electrons.

Higher Order Modes (HOMs)



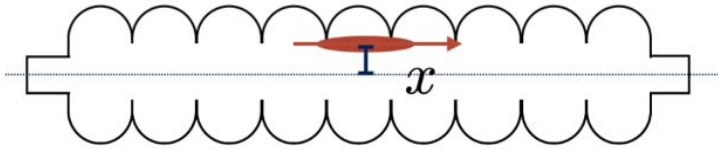
PHYS. REV. ACCEL. BEAMS 25, 064402 (2022)
HOM waveforms for cavities 1–8

Figure 1: 3.9 GHz third harmonic cavity.



https://accelconf.web.cern.ch/SRF2015/papers/thpb008.pdf

$$V_x(t) \propto x \cdot e^{-(t/2\tau)} \sin(\omega t),$$



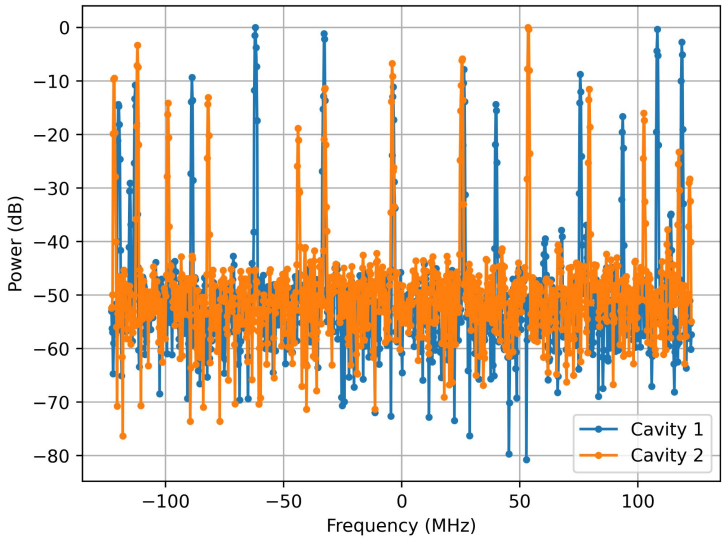
PHYS. REV. ACCEL. BEAMS 20, 123501 (2017)

Qualitative Description

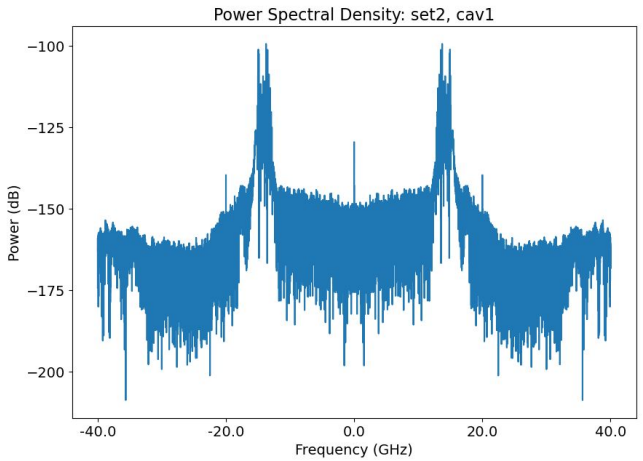
- When the electron beam is misaligned, it loses energy and spreads out.
- This leads to “high emittance”—transverse, cross-sectional area occupied by the beam.
- Electron bunch loses energy in the form of electromagnetic waves—can be expanded as “eigenmodes,” with the higher order modes (HOMs) leading to beam breakup and instability.
- HOM couplers, antennae attached to each cavity, dissipate and monitor HOM frequencies which are processed and archived.
- HOM signal amplitude is proportional to the charge of the bunch, cavity axis offset, trajectory tilt, and bunch tilt while on axis.
- Other parameters are the frequency and decay time of the dipole mode.

Higher Order Modes Signal Processing

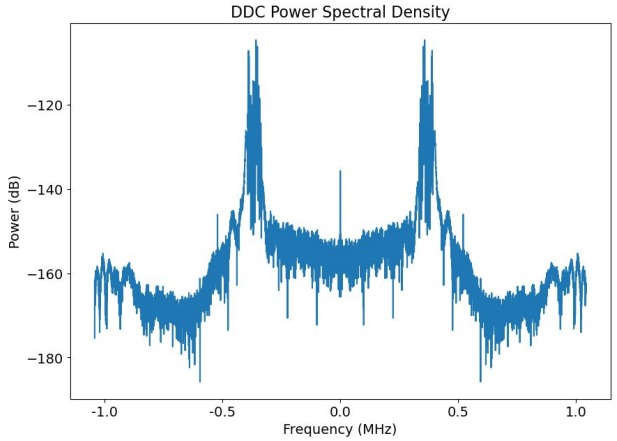
Chao's FFT



Justin/My PSD, DDC



https://pysdr.org/content/sampling.html#calculating-power-spectral-density



Qualitative Description

- Two methods explored: Chao's ZCU216 evaluation board and Justin's Fast Scope.
- Chao's board digitizes the rf signals, and performs down conversion.
- Chao also performed a fast fourier transform to isolate the peak frequencies from the data.
- Justin's oscilloscope does not automatically do this, and I attempted to do a DDC.
- Power Spectral Density: average power by taking the magnitude of each sample, squaring it, and then finding the mean
- Digital Downconversion:
  - Filtering - remove unwanted spectral components, prevent aliasing
  - Mixing - multiply the signal to shift the frequency spectrum
  - Decimation - shift sampling rate to match the new, lower frequency

Motivation and Strategy

Project Description and Goals

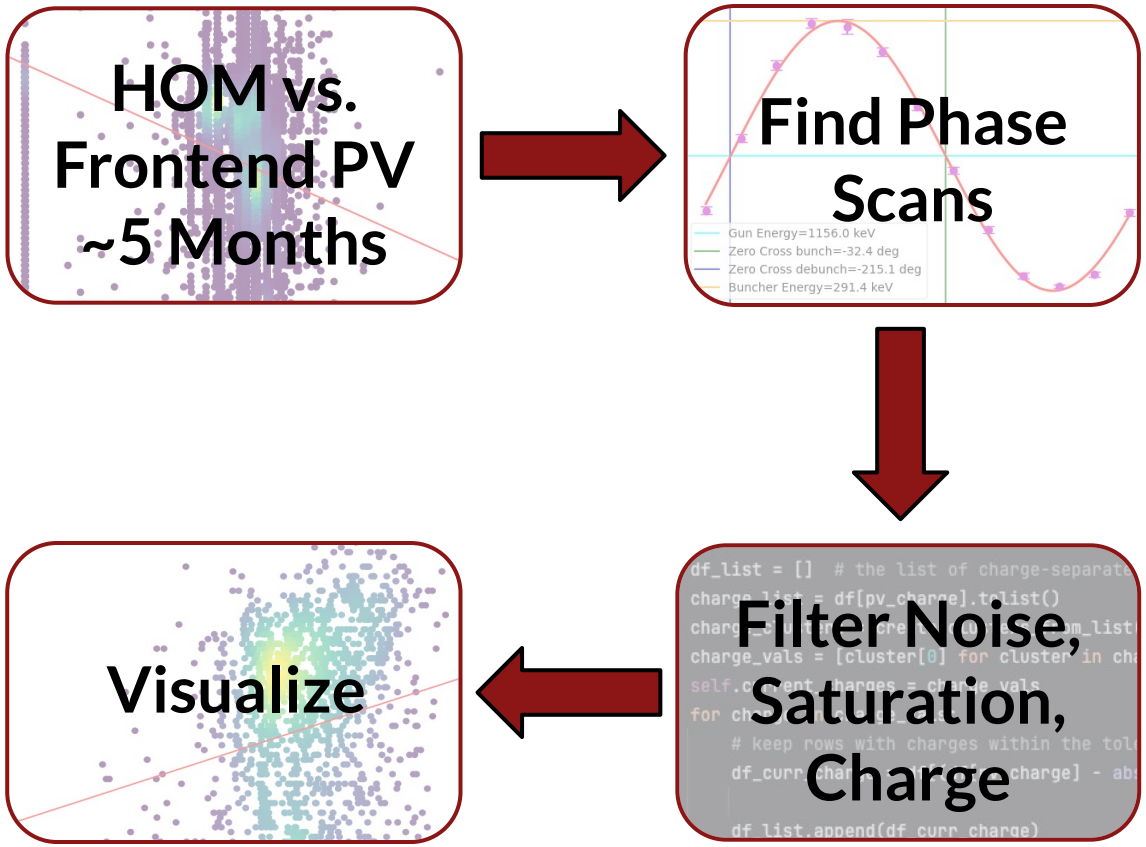
Higher Order Mode signals, bunch charge, steering magnet strength, and beam position/offset are used to create a function that gives precise information on **how beam line components can affect the beam dynamics.**

To achieve this, one must:

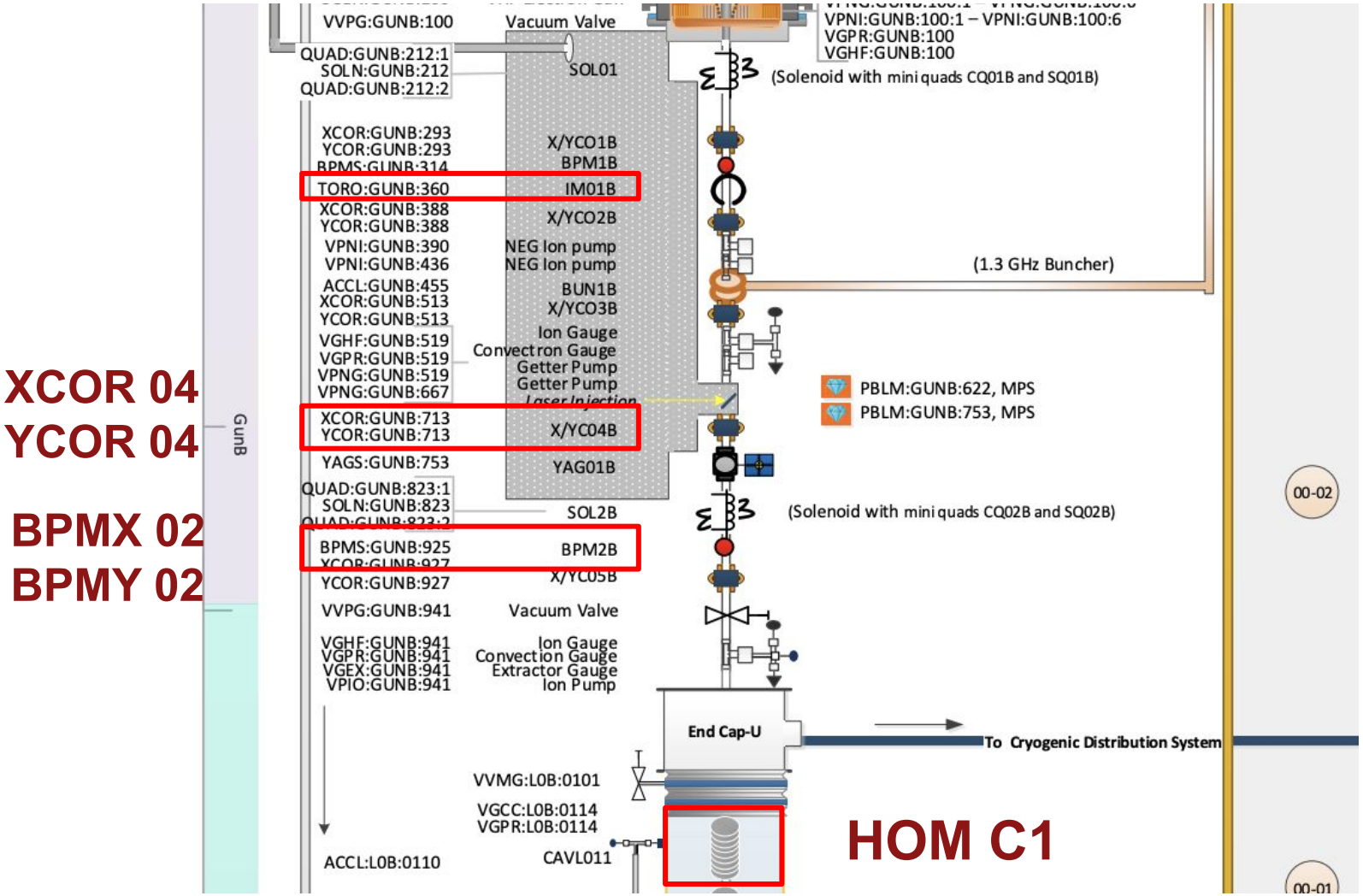
- Analyze and process months of archived data (archiver.py, EPICS, MEME).
- Determine which correlations and PVs are relevant to this question.
- Develop algorithms to efficiently access, process, and display data.

Long-term goal to develop an algorithm that will:

- enable seamless control over the beam.
- reduce energy loss and emittance.
- enable the LCLS-II to reliably generate up to 1 million. X-ray pulses per second with minimum loss to HOMs.



LCLS-II



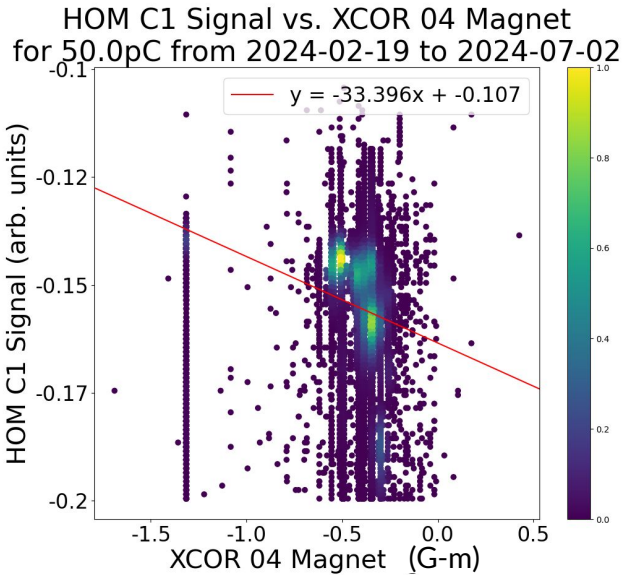
Beamline

- Top-down view.
- Shows key components that align, monitor, and focus the beam.
- **TORO:GUNB:360:CHRG** measures bulk charge of the beam in pC.
- **XCOR/YCOR:** correcting magnets that tune the beam in respective transverse directions.
- **BPMS:** “beam position monitor” that measures in the X and Y transverse positions.
- Not listed: **SCOP:AMRF:RF01:AI_MEAS1** with readout of HOM signals.
- Data is archived and can be accessed in multiple ways.

Summary (5 Month, 300k pts.)

- Visualized HOM C1 Signal vs. XCOR 04/YCOR 04/BPM 02 over the past 5 months.
  - Cluster of high HOM signals when XCOR 04 at around -0.3—0.5 G-m.
  - Higher HOM signals correlate directly with greater YCOR 04 strength.
  - For both a greater X and Y position offset, the trend predicts a lower HOM signal—the opposite of what we expect.
  - Upon closer observation: clusters of points around -5 mm for the BPM X and -4 mm for BPM Y.
    - Indicates either a consistently off-axis beam or an off-centered BPM location.

1



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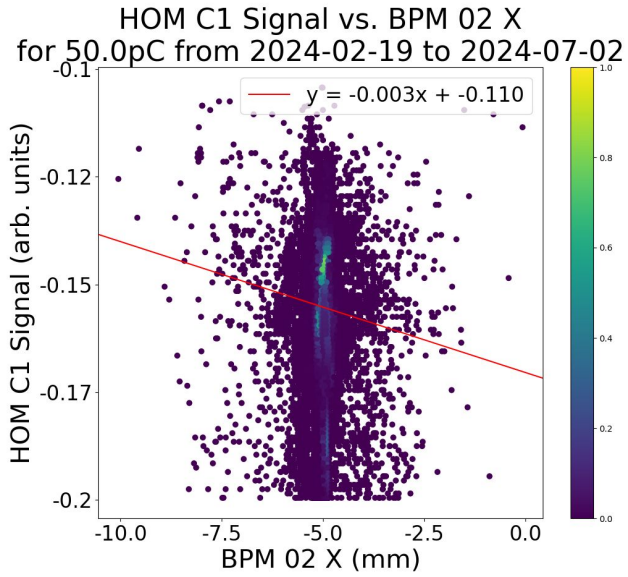

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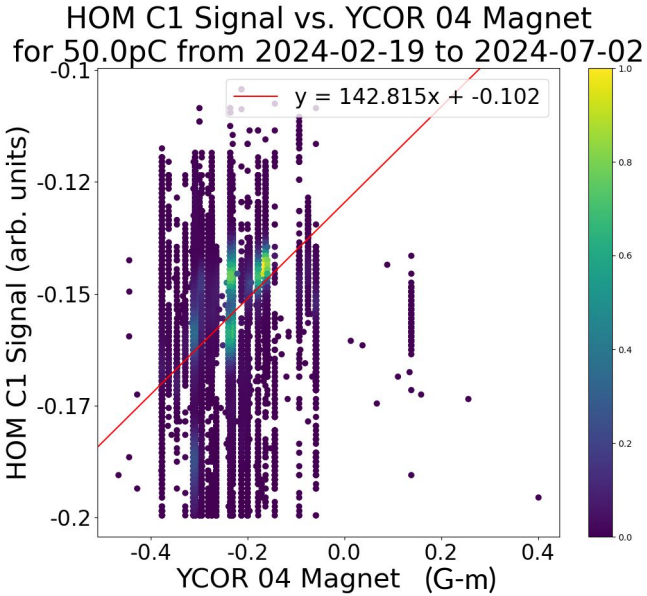

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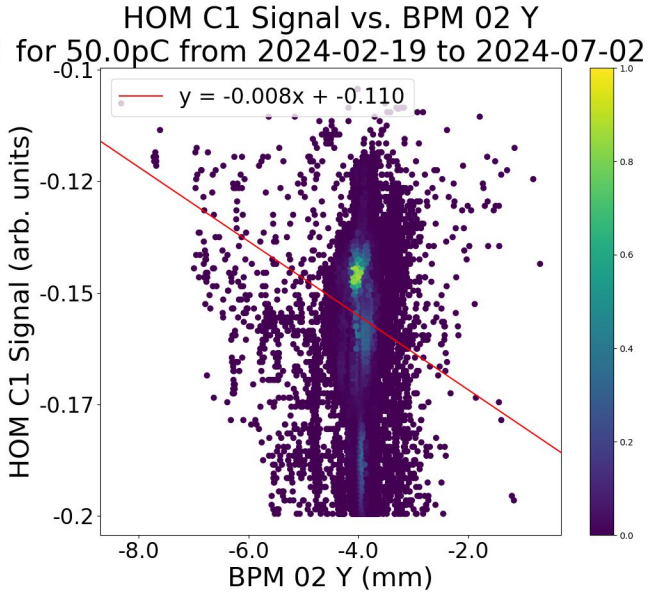
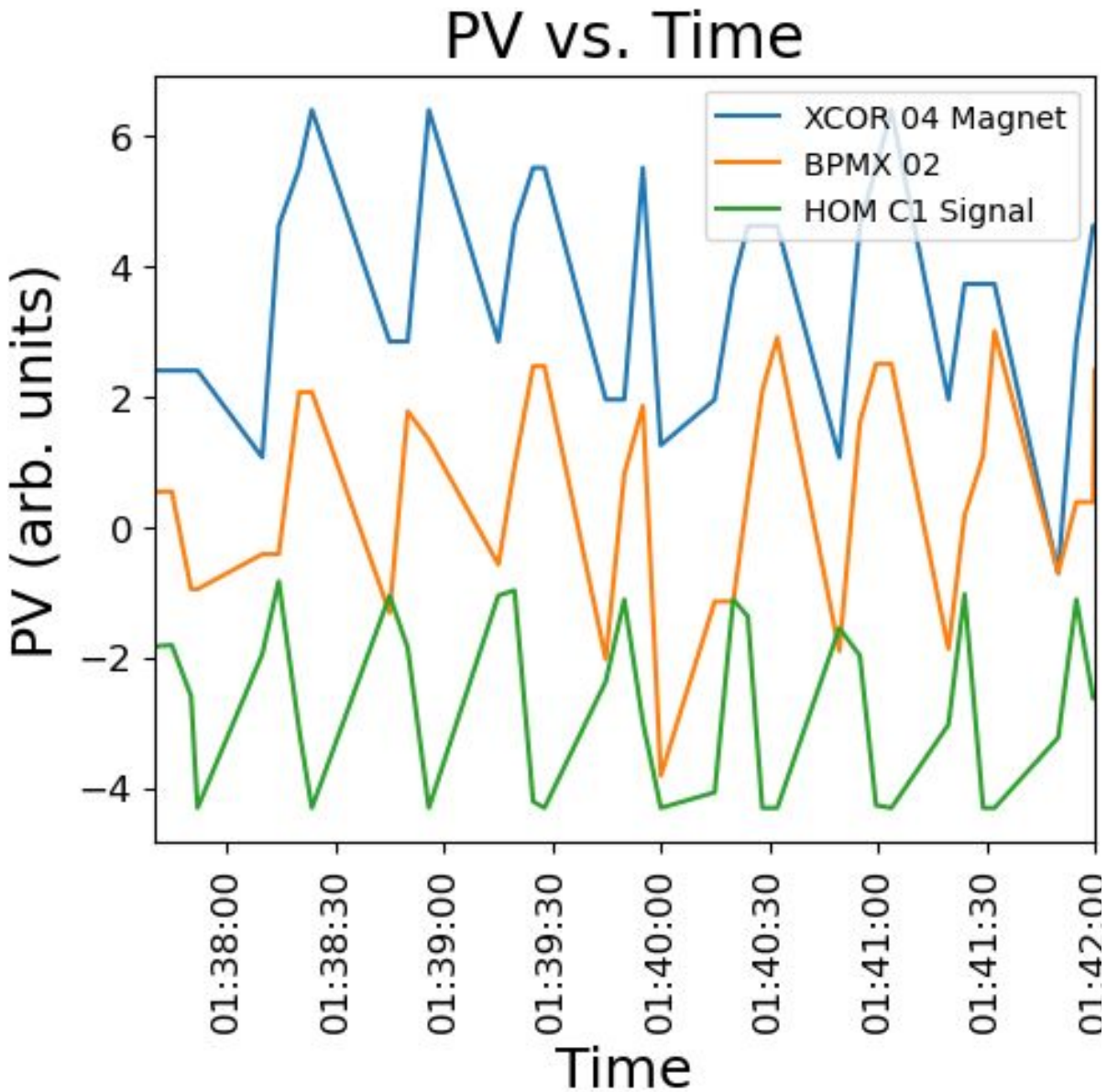
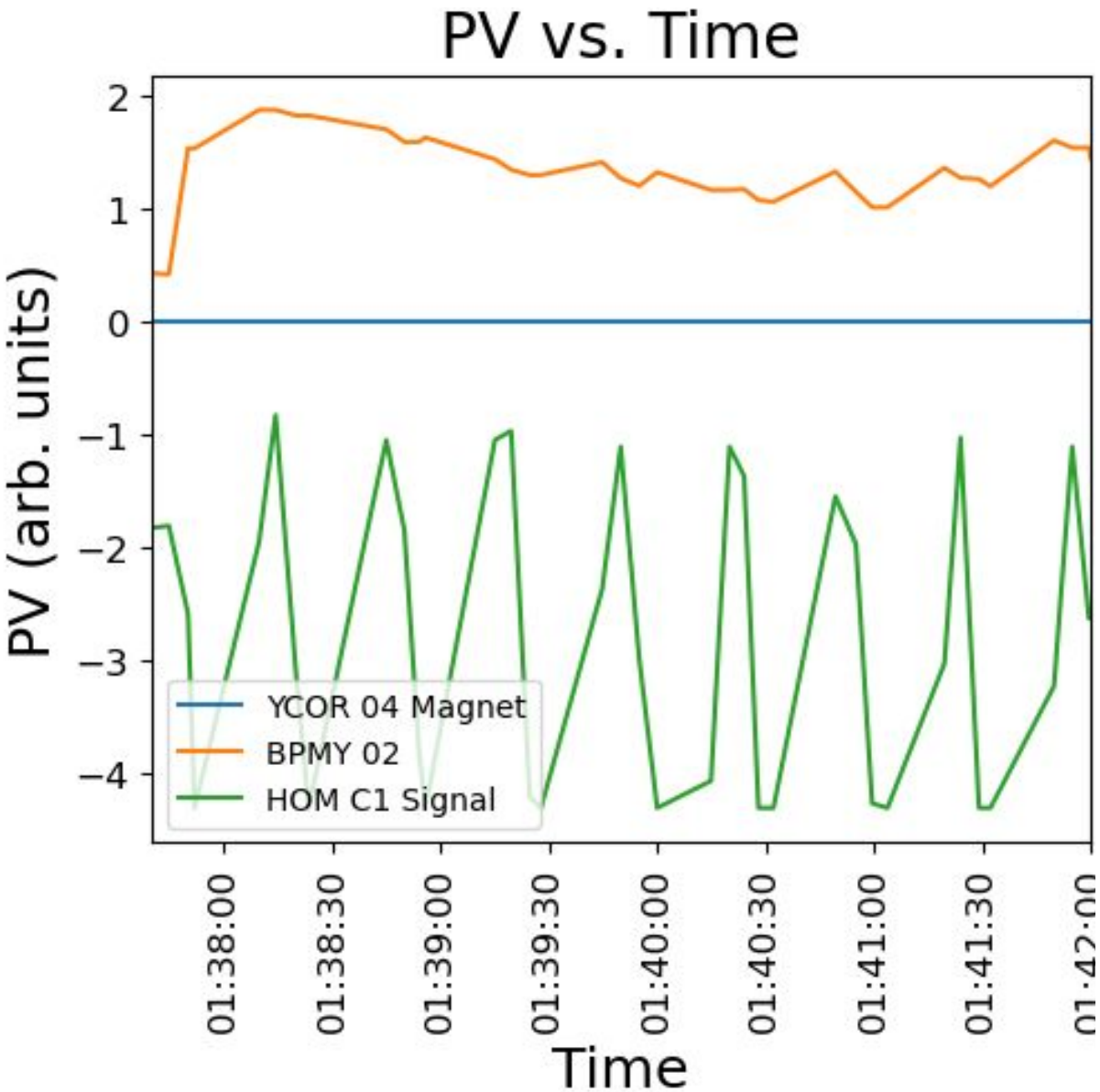


Injector Buncher Phase Scans: June 7th, 2024

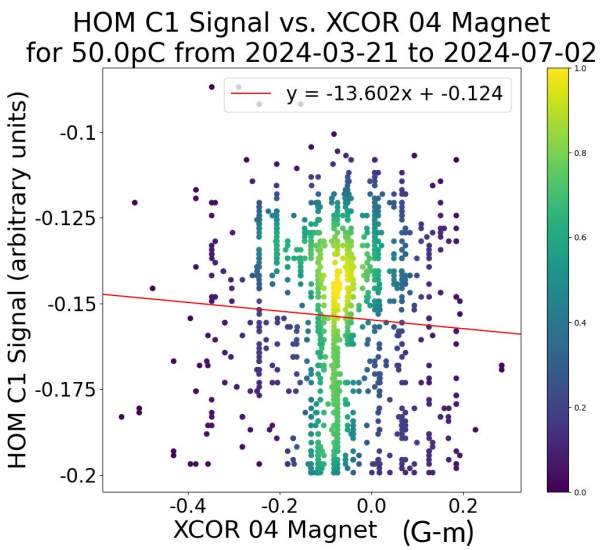




Summary (Phase Scans, 1k pts.)

- Again, difficult to discern clear trends due to overabundance of data, despite isolating phase scans.
  - Cluster of -0.15 HOM signals at around -0.1 G-m for the XCOR and -0.15 G-m for the YCOR.
  - Clearer trend for YCOR indicating higher magnet strengths correlating with larger HOM signals.
  - Clusters of -0.15 HOM signals at -4.0 mm for BPM X and -2.0 mm for BPM Y.
    - Again, indicates either a consistently off-axis beam or an off-centered BPM location.
  - Cause of off-axis beam could be the result of other PVs following the buncher, not just XCOR/YCOR 04.

1



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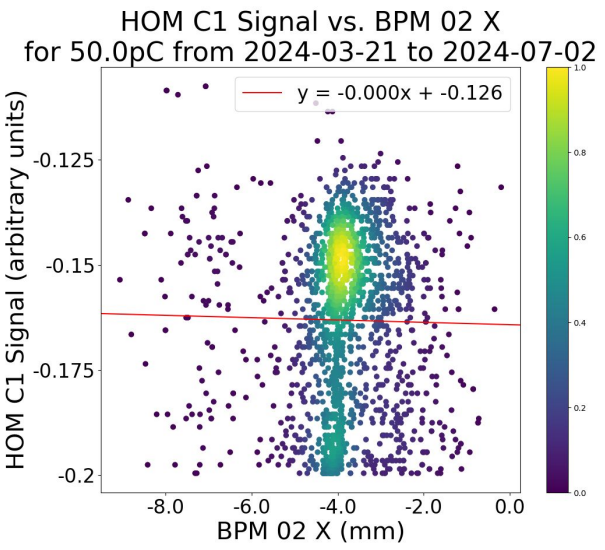

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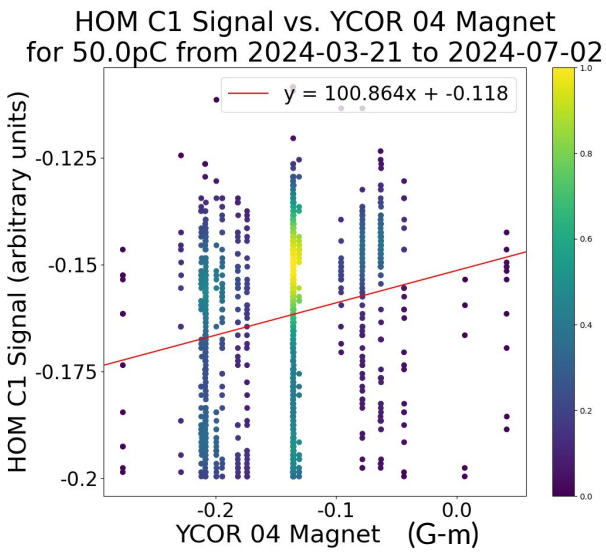

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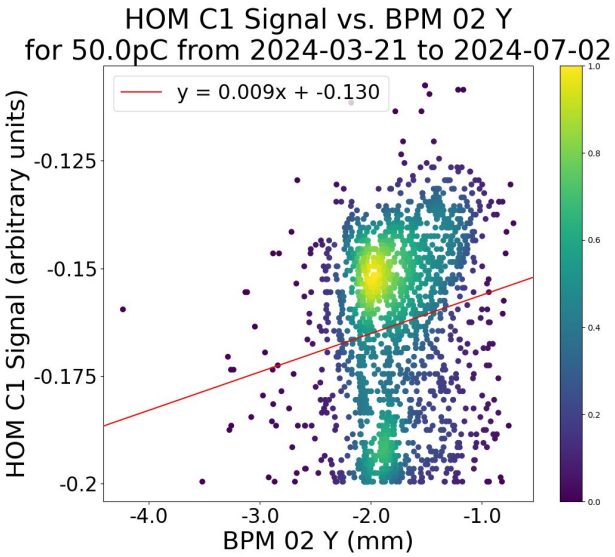
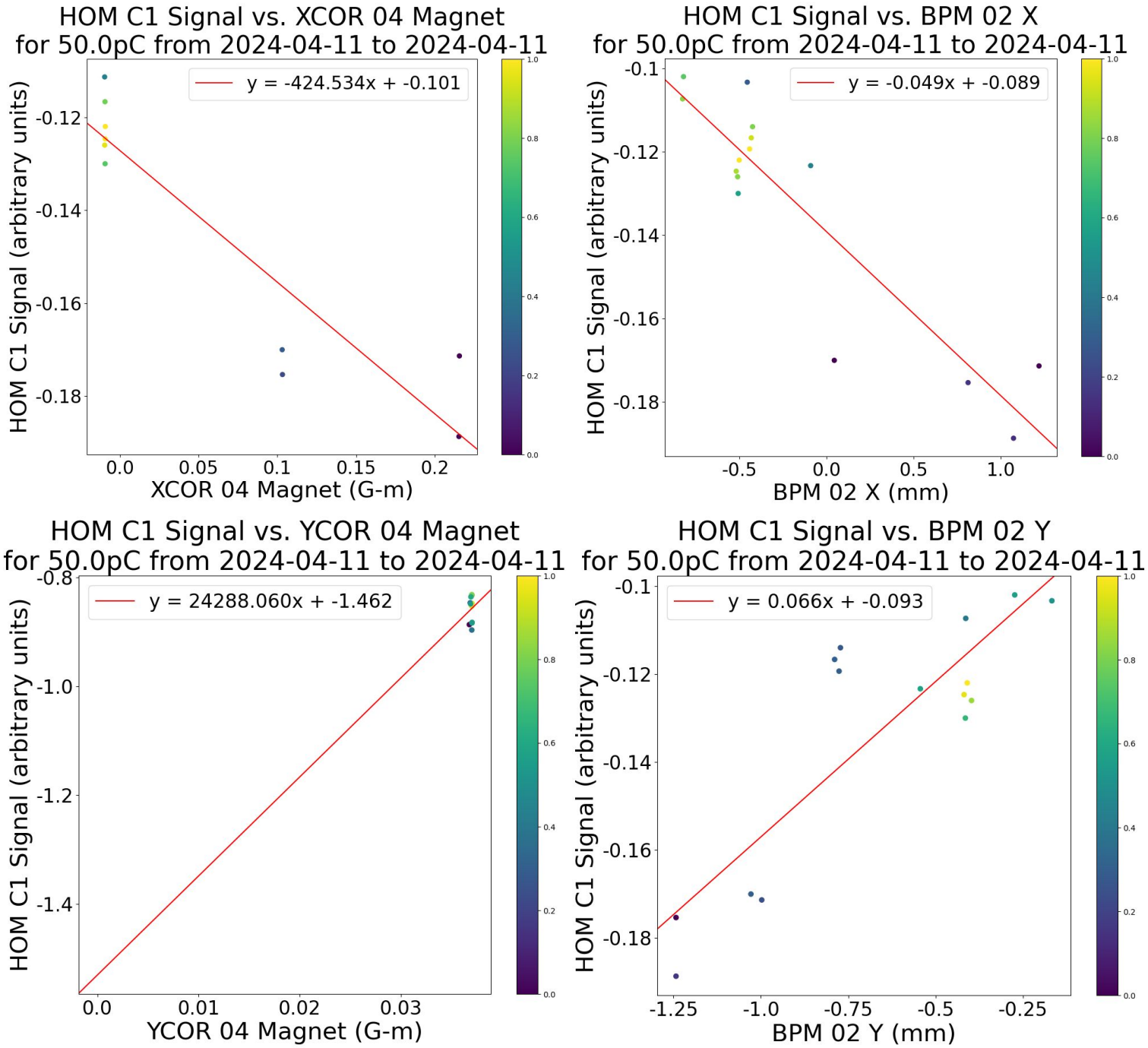
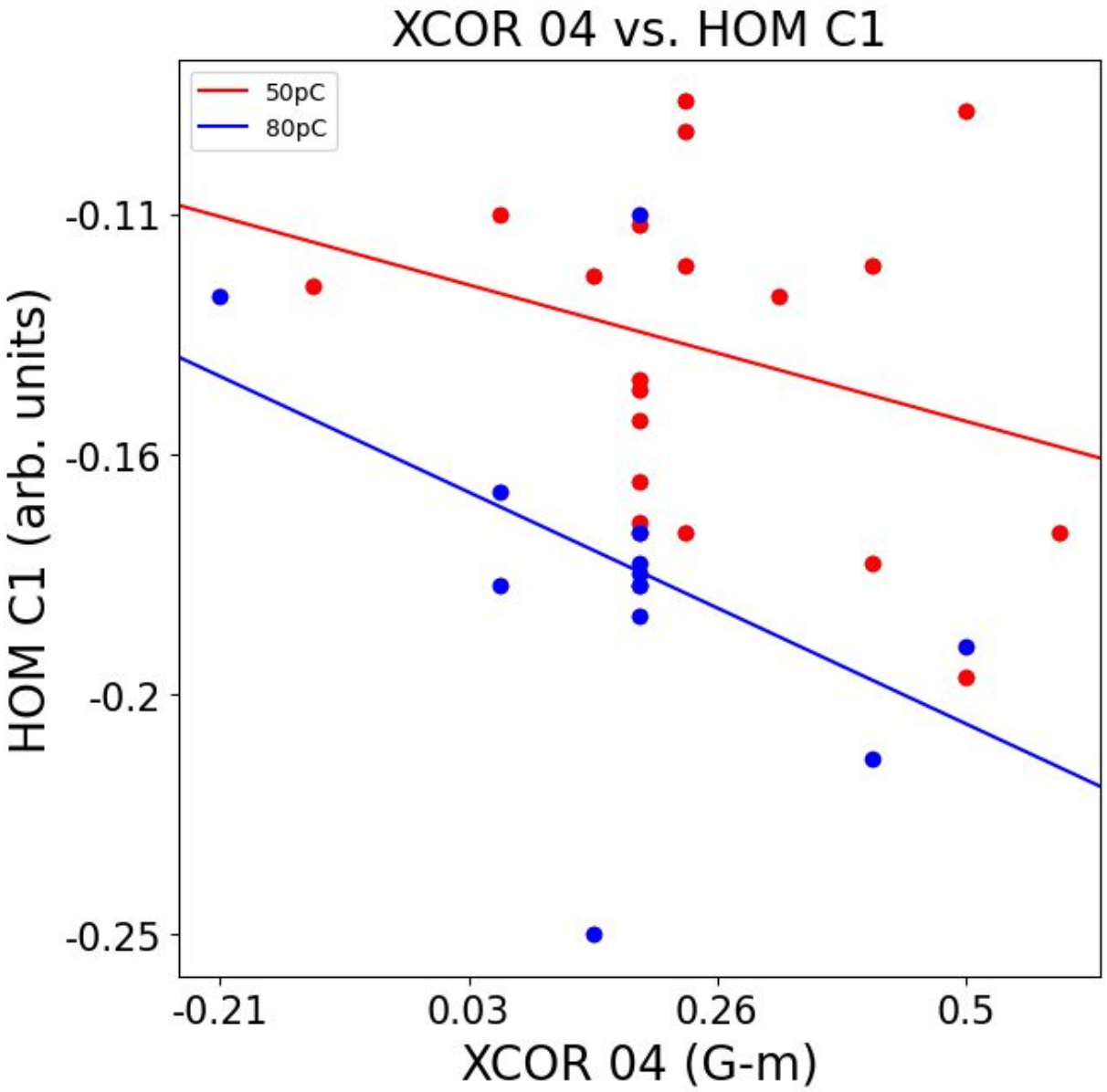
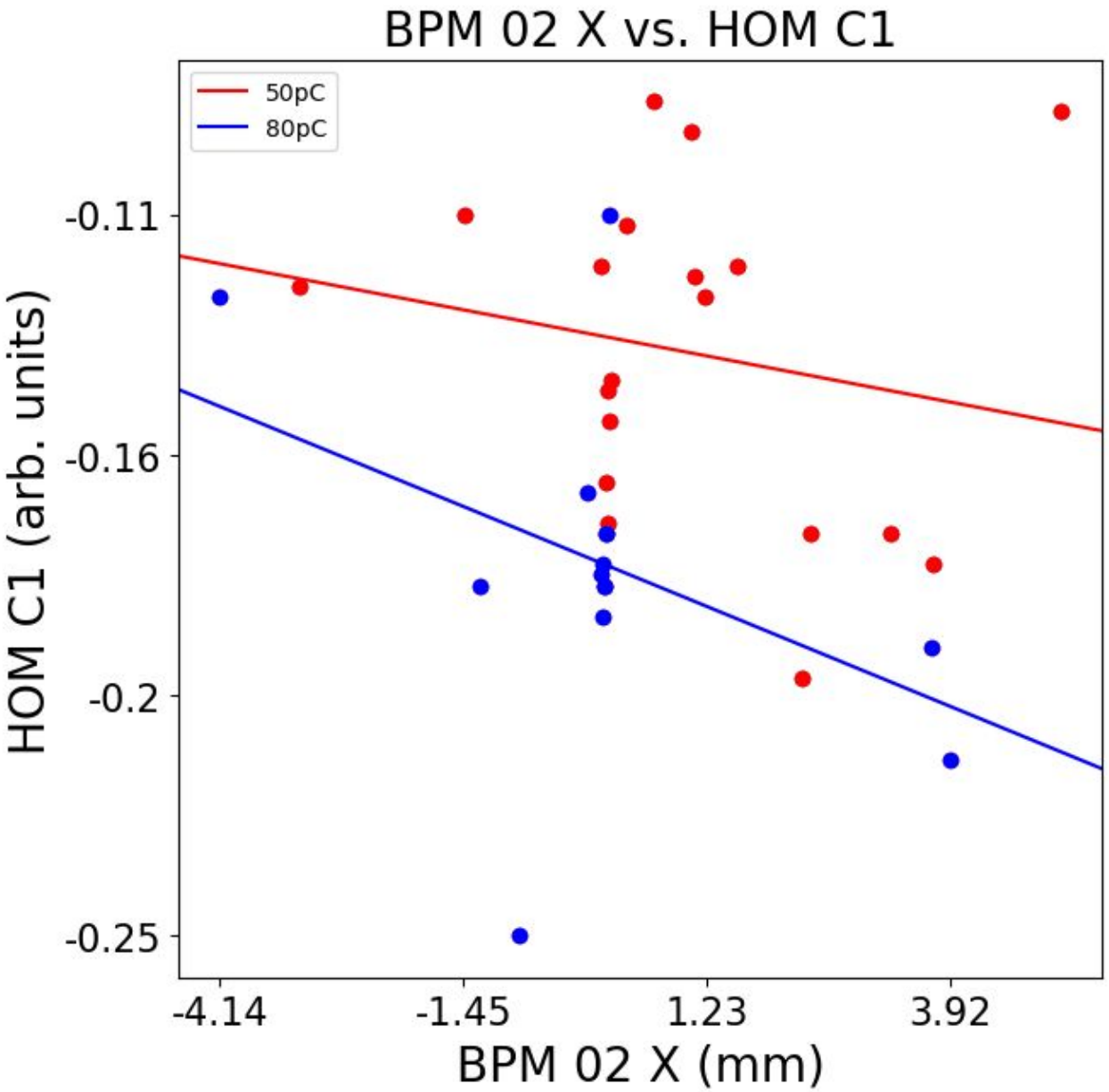


Phase Scan (set #9) 50pc



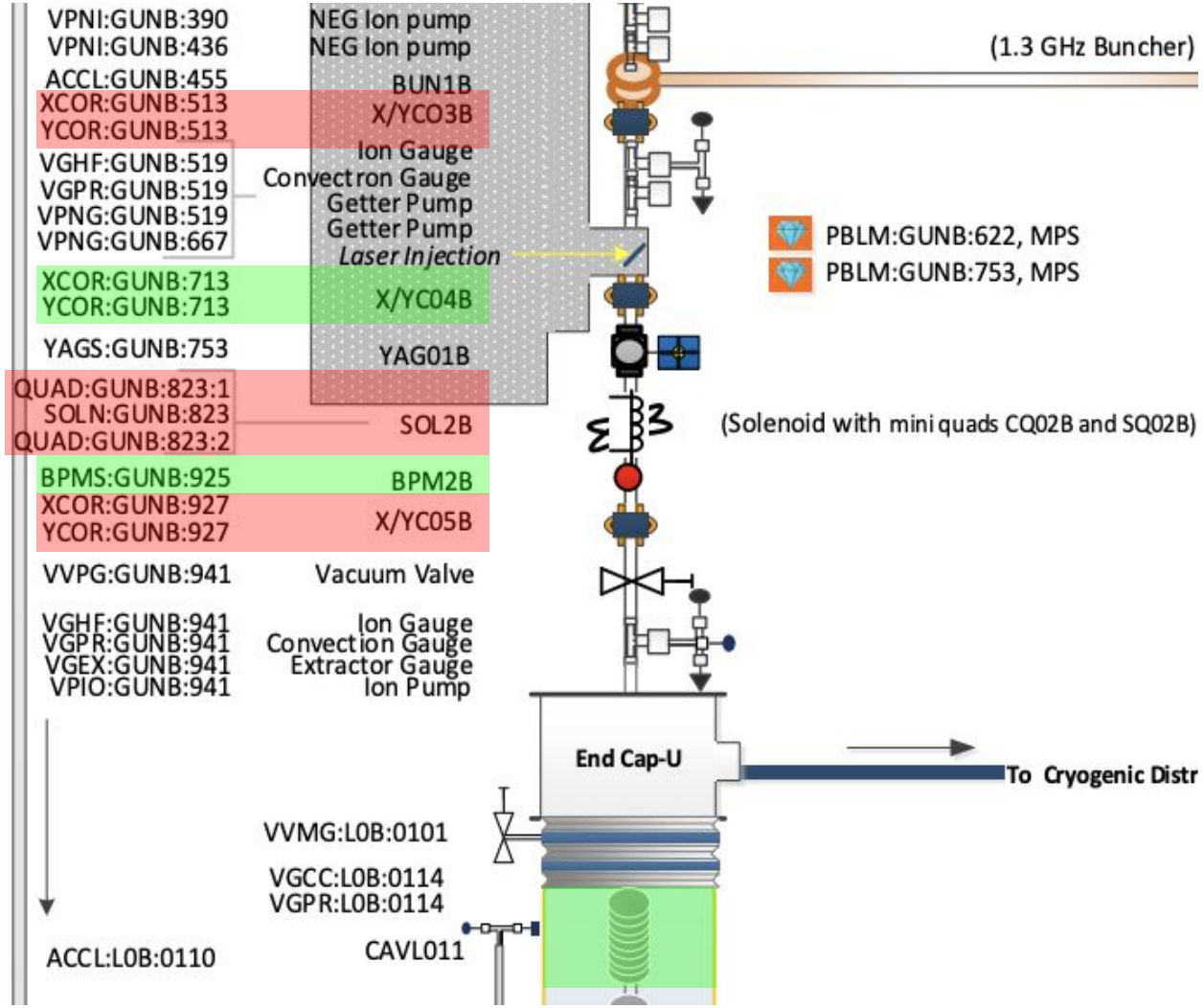
Phase Scan (set #76) 50pC and 80pC





Conclusion

- Farther off-axis beams in the y transverse position correlate with stronger HOM signals.
- The beam is either off-axis coming into BPM 02, or that may simply be where the BPM is located in the tube.
- For certain phase scans, there is more data for specific charge values, especially 50pC.
- Difficult to discern trends over multiple phase scans, unless they happen to be very similar in design.
- Future goals are to more carefully isolate similar phase scans for specific correcting magnets and bunch charges.
- There may be additional steering besides X/YCOR 04.
    - Other PVs following the buncher could give better insight.
- One step closer to a unified algorithm describing how individual PVs impact the beam offset and HOM signals.



Thank you for your attention

Questions?

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Thank you to Dr. Nicole Neveu for providing great mentorship throughout my time as a SULI Intern.